EXPERIMENT 12

Port Scanning Tools & Techniques (SYN, TCP Connect, UDP)

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AIM:

To execute and demonstrate the functionality, syntax, and outputs of various port scanning techniques and tools in Kali Linux Operating System, including SYN Stealth Scan (-sS), TCP Connect Scan (-sT), and UDP Port Scan (-sU).

COMMANDS AND DESCRIPTION:

1. TCP SYN Stealth Scan (nmap -sS):

• *Description:* The default and most popular scan type in Nmap. It sends a SYN packet and waits for a SYN/ACK (open) or RST (closed). Because the full three-way handshake is never completed, it is fast and stealthy.
• *Syntax:* sudo nmap -sS [options] <target_IP>
• *Privilege:* Requires raw packet privileges (sudo/root).

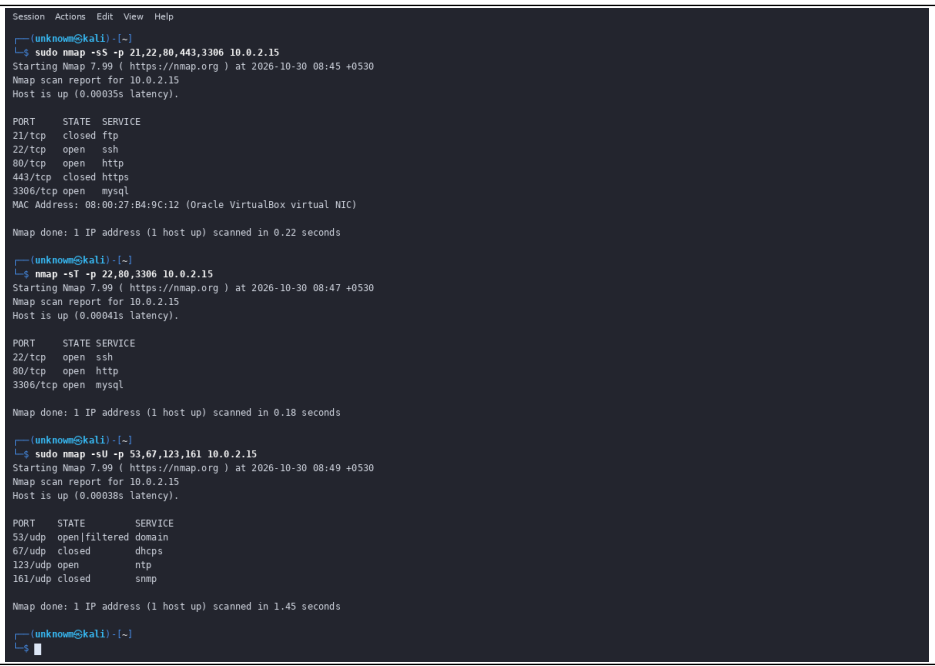
2. TCP Connect Scan (nmap -sT):

• *Description:* Completes the full three-way TCP handshake (SYN, SYN/ACK, ACK) using the standard operating system network socket API. Used by unprivileged users when raw packet access is unavailable.
• *Syntax:* nmap -sT [options] <target_IP>
• *Privilege:* Can be executed without root privileges.

3. UDP Port Scan (nmap -sU):

• *Description:* Scans for open UDP services (such as DNS on port 53, DHCP on port 67, NTP on port 123, SNMP on port 161) by sending empty or service-specific UDP packets and listening for ICMP Port Unreachable responses.
• *Syntax:* sudo nmap -sU [options] <target_IP>
• *Characteristics:* Slower than TCP scanning due to RFC 1812 ICMP rate limiting on modern operating systems.

OUTPUT:



RESULT:

Thus, port scanning tools and techniques including TCP SYN stealth scan (-sS), TCP connect scan (-sT), and UDP port scan (-sU) were successfully executed on the target host in Kali Linux Operating System, verifying open ports, protocol states, and service availability.